SOFTWARE REQUIREMENTS DOCUMENT

Life.Trace | Private Backtracker v1.0

Project Name	Life.Trace(Private Backtracker)
Version	1.0
Date	June 15, 2026
Author	For personal use
Classification	Private — Not for Distribution

⚠ **Strictly for personal research and private use only**

1. Introduction

1.1 Purpose

Build a personal, offline-capable or lightly online tool that aggregates publicly available information for personal research — e.g., verifying contacts, finding lost people, and performing reverse lookups. This tool is strictly for your own private use and is not intended for public distribution.

1.2 Scope

In-Scope:

- Reverse image search across multiple engines
- Username lookup across 400+ platforms
- Phone number lookup and carrier identification
- Name-based web mentions and people search
- Result organisation, dashboard view, and export

Out-of-Scope:

- Hacking or accessing private accounts
- Accessing paid or premium databases without permission
- Storing data on public or third-party servers
- Any commercial use or distribution

1.3 Definitions & Acronyms

Term	Definition
OSINT	Open Source Intelligence — gathering data exclusively from publicly available sources
MVP	Minimum Viable Product — the core feature set needed for initial release
Reverse Image	Finding where a photo appears online by using the image itself as a search query
VPN	Virtual Private Network — used to mask your IP and bypass rate limits
SRD	Software Requirements Document — this document

2. Overall Description

2.1 Product Perspective

A simple personal dashboard that combines multiple free and open-source OSINT tools into one interface, reducing the need for manual tab-switching and repetitive searches. The goal is a unified experience where results from one module can feed into another.

2.2 User Characteristics

- Single user — you (located in Pakistan)
- Technical level: Beginner to intermediate Python user
- Expected use: periodic personal research, not continuous monitoring

2.3 Operating Environment

- Desktop or Laptop — Windows 10/11 or Linux (Ubuntu/Debian preferred)
- Python 3.10 or later
- Reliable internet connection (VPN strongly recommended for privacy and avoiding rate limits)
- All data stored locally — no cloud dependency

3. Functional Requirements

3.1 User Inputs

The system shall accept the following input types from the user:

- Image file upload (JPG, PNG) — for reverse image search
- Phone number with country code — e.g., +92 300 1234567
- Full name, with optional city or country for scoping
- Username or social media handle
- Email address (optional, for additional correlation)

3.2 Core Features (MVP)

The following table defines the minimum viable feature set and their priorities:

Module	Priority	Description
Reverse Image Search	**High**	Upload photo → search Yandex, Google Lens, TinEye, Bing. Display matching images and source links.
Username Search	**High**	Input username → check 400+ platforms using Sherlock / WhatsMyName. List active accounts with direct links.
Phone Number Lookup	**Medium-High**	Input number → use PhoneInfoga and public sources. Show carrier, linked accounts, and any public mentions.
Name Search	**Medium**	Input name → run Google dorks and public people-search aggregators. Summarise results with source links.

Module	Priority	Description
Result Dashboard	**High**	Unified tabbed view for each input type. Cross-link related results (e.g., image → found username).
Export	**Medium**	Save results as PDF, HTML, or a JSON folder for archiving.
Search History	**Low**	Local log of past searches, stored in SQLite, searchable by date or keyword.

3.3 AI Assistance (Optional Enhancement)

The following AI-powered features may be added post-MVP:

- Summarise long or cluttered OSINT results into a readable brief
- Suggest logical next search steps (e.g., "This username also appears on Instagram — review photos?")
- Highlight patterns or links across multiple result sets automatically

4. Non-Functional Requirements

4.1 Performance

- Individual searches should complete within 60–90 seconds where possible
- Username platform checks may take longer depending on network speed and proxy use

4.2 Security & Privacy

- All search results stored locally — zero telemetry or cloud uploads
- Proxy and VPN support built in to reduce exposure of your IP address
- No third-party analytics or tracking libraries

4.3 Usability

- Simple graphical interface using Streamlit or Gradio — no command-line knowledge required for day-to-day use
- Clear labelling of result sources so you can manually verify accuracy

4.4 Reliability

- Graceful handling of rate limits — automatic delays and retries
- Blocked-site detection with informative error messages, not crashes

4.5 Maintainability

- Modular Python codebase — each OSINT module is a separate file

- Clear inline comments throughout the code
- A simple config file for updating API keys, proxies, and platform lists

5. Technical Constraints & Recommendations

5.1 Language & Runtime

- Primary language: Python 3.10+
- Node.js may be used for Electron-based GUI if preferred over Streamlit

5.2 Key Libraries & Tools

Library / Tool	Purpose
Sherlock / WhatsMyName	Username search across 400+ platforms
PhoneInfoga	Phone number OSINT — carrier, location, linked accounts
Playwright / Selenium	Browser automation for Yandex, Google Lens, etc.
Streamlit or Gradio	Simple web-based GUI dashboard
Requests + BeautifulSoup	Web scraping and HTML parsing
SQLite	Local storage for history and structured results
ReportLab / WeasyPrint	PDF export of results

5.3 Data Storage

- SQLite database for search history, tags, and result metadata
- Flat folder structure (JSON + HTML) for raw result exports
- No external database required — fully local

5.4 Pakistan-Specific Considerations

- Support for common Urdu-transliterated name variations in search queries
- Country-code aware phone parsing (+92 prefix handling)
- Proxy support tested against Pakistani ISP restrictions

6. Assumptions & Dependencies

- You have basic Python knowledge or are willing to follow setup guides
- A stable internet connection is available during searches
- You will use a VPN for sensitive searches to avoid tracking or rate limits
- All tools used respect platform Terms of Service — only public data is accessed

- The tool will not be shared, published, or used for commercial purposes

7. Risks & Mitigations

Risk	Mitigation
Rate limiting or IP blocks from platforms	Use configurable delays between requests, rotating proxies, and VPN. Display informative messages rather than crashing.
Inaccurate or stale OSINT results	Always display the source URL so results can be manually verified. Do not treat results as definitive ground truth.
Legal or ethical exposure	Keep the tool entirely private and personal. Only access publicly available information. Do not store or share third-party personal data.
Tool/library deprecation	Use modular code so any individual integration can be swapped. Pin dependency versions in requirements.txt.
False positives in username matching	Display confidence indicators where possible. Cross-reference results across multiple modules before drawing conclusions.